Fluid Automation
Laboratory report

Group 08

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Politecnico di Torino

Master Degree in Mechatronics Enginnering

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Fluid Automation

Task 1

Piece-counting system with a conveyor belt

Explanation

Conveyor belt driven by a **pneumatic actuator** through a **start** and a stop **button**

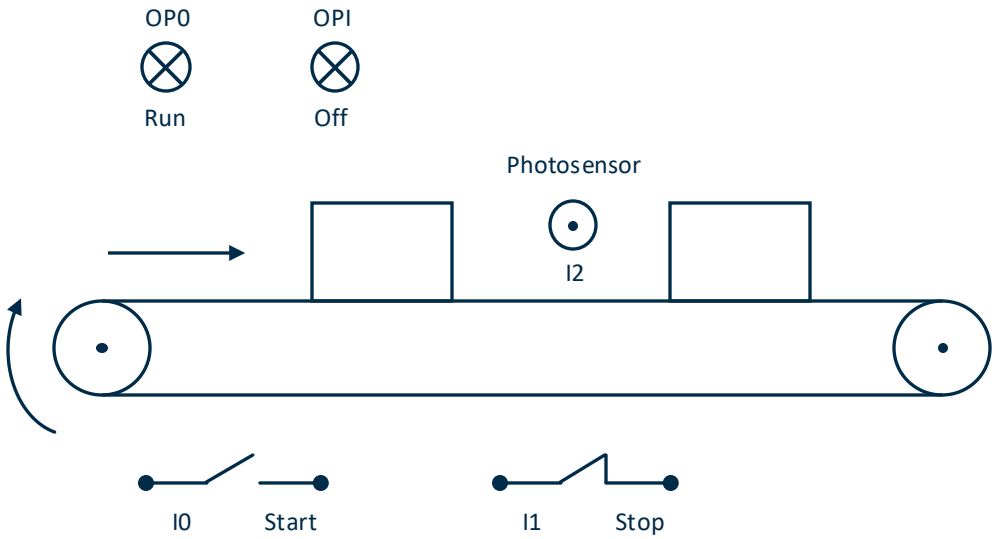
Start and **stop** status are detected by two distinct **lamps**

Pieces to be counted are placed on the belt

Each time one of the pieces passes in front of the **photocell** it is **counted**

The belt can be **stopped** at any time by pressing the **stop button**

The belt can be **restarted** by pressing the **start** button again, piece counting is **resumed** from where it was interrupted



Objectives

AUTOMATIC CONTROL

- Automation Studio
- Ladder code
- Grafcet code

PNEUMATICS

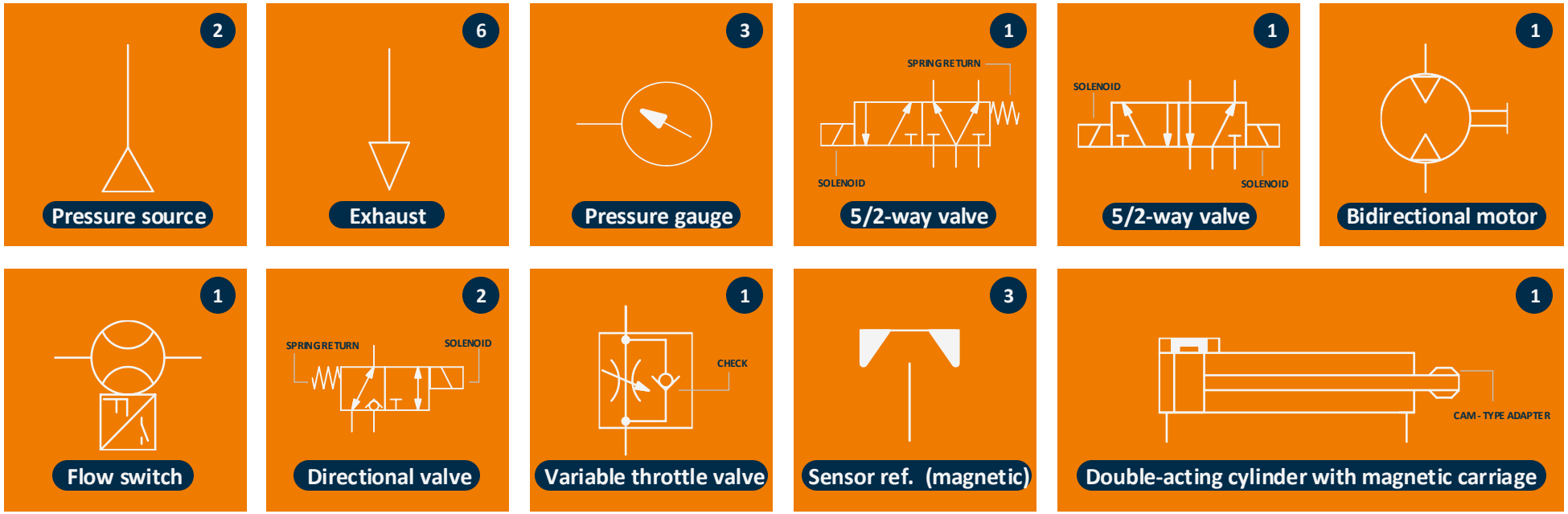
- Pneumatic cyliders
- Electrovalves

HMI INTERFACES

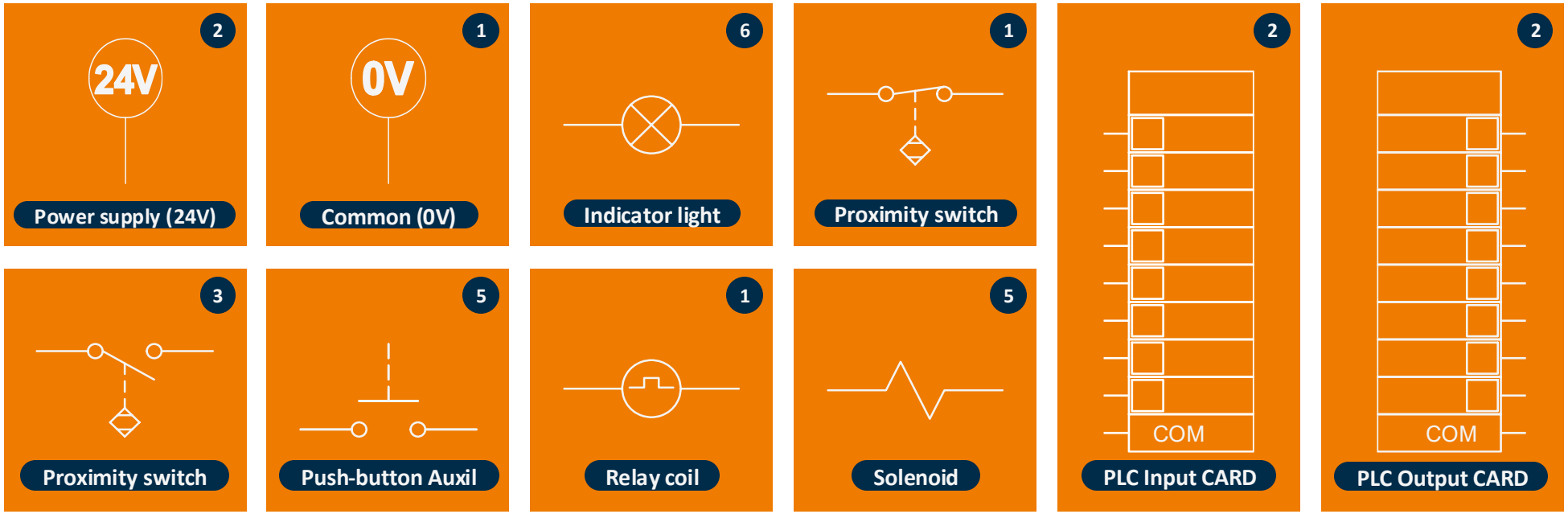
- Pushbuttons
- Timers
- Actuators
- Sensors
- Lights
- Alarms



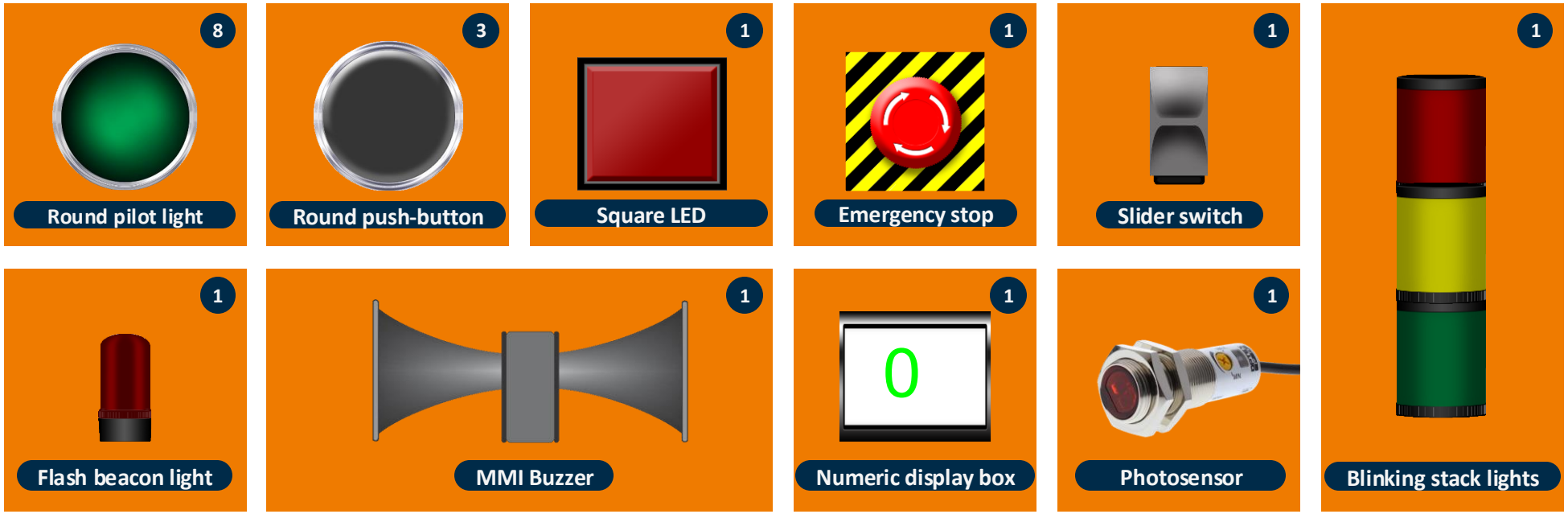
Components pneumatic



Components electrical



Components HMI



Grafcet

Start/Stop memory setting

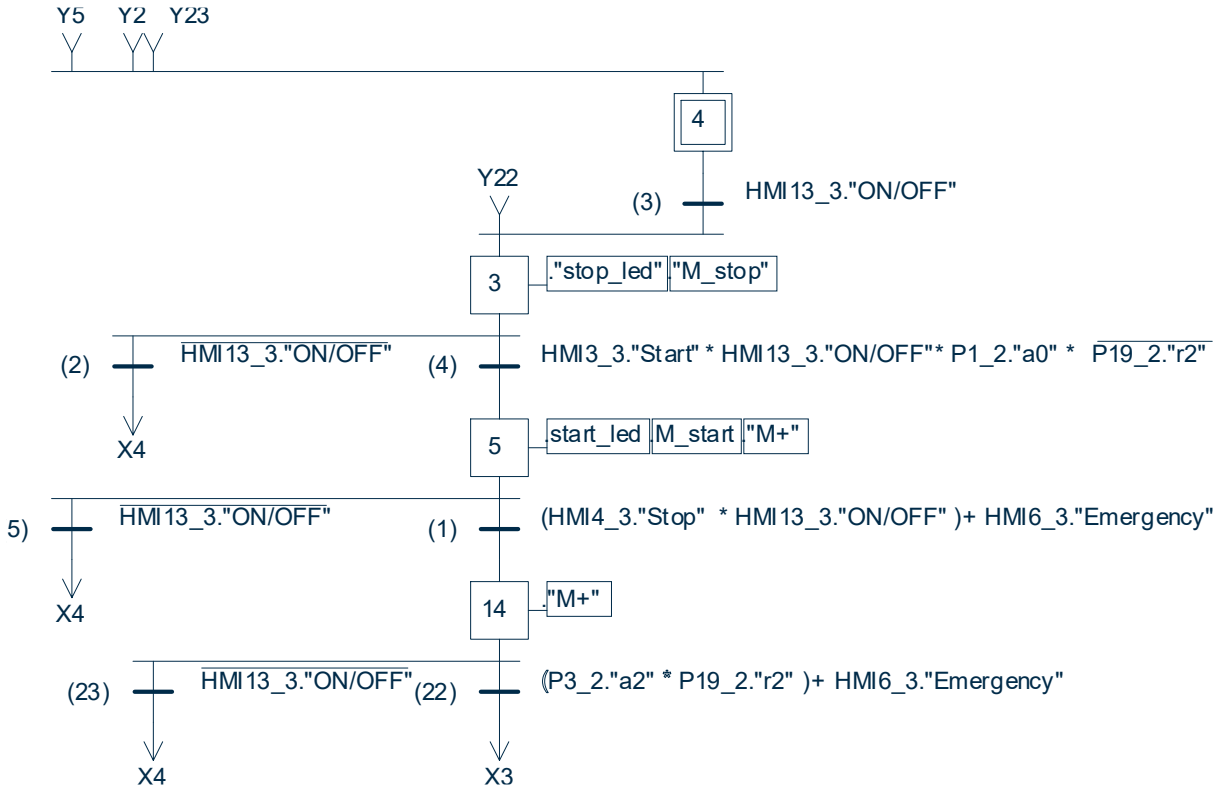
When the **ON/OFF** switch toggles, the **stop LED** and **memory** are activated

If the **start** push button is pressed, the **start memory**, **LEDs**, solenoid **M+** are turned **on**

If the **stop** push button is pressed, the **piston** arrives at the position **a2**, **retracts** and **stop** and the LEDs are turned on

At each a stage **control** is performed on the **ON/OFF** activation and, if **true**, the cycle is brought back to the **initial step**

At each start **control** is performed on the **emergency** activation and, if **true**, the cycle is interrupted, and **stop** is set



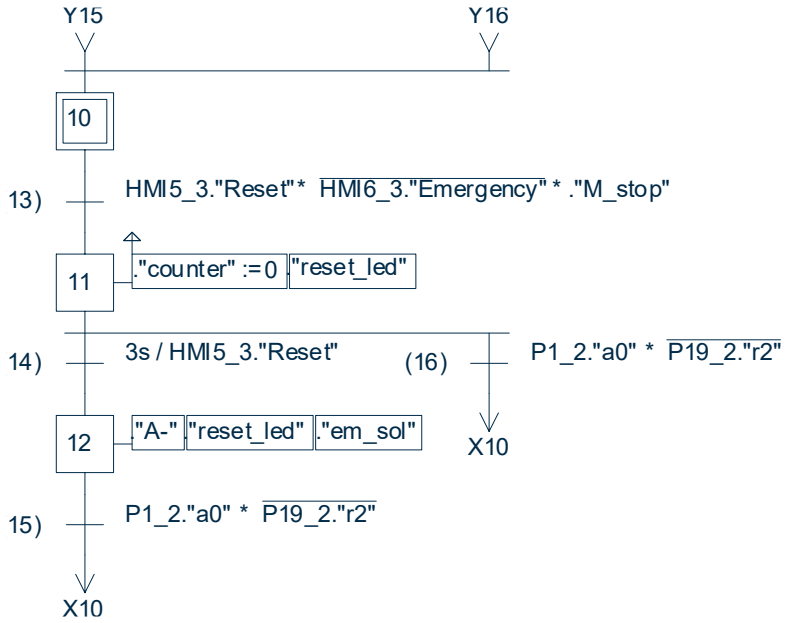
Grafcet

Reset

If the **reset** button is pressed, the **stop memory** is set, and the **emergency** is not pressed, the counter is **reset** and the **reset LEDs** switch on.

HOLD 3s

If the **reset** button is pressed for **3s** the piston is retracted to **a0** and the cycle **restart** from the initial step; if the piston is already in position **a0**, the cycle simply restart from the initial step



Grafcet

Start cycle

When the **start** push button is pressed and the piston is at its initial position, the outward stroke begins (***A+***)

a1

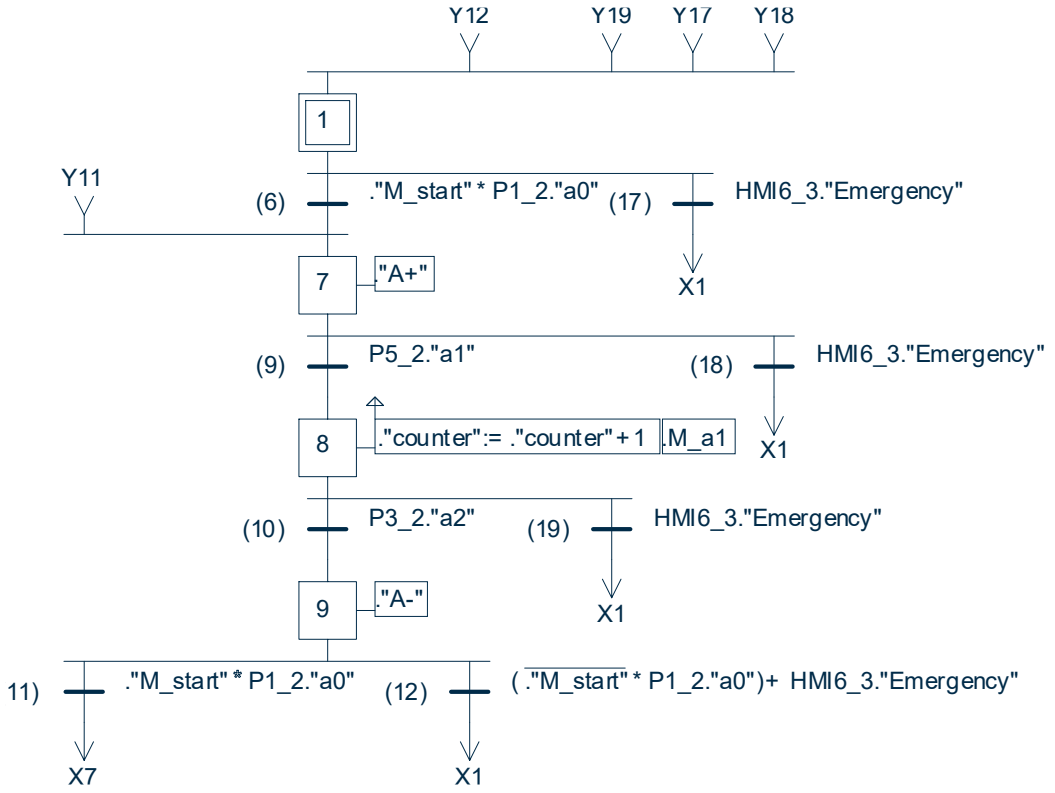
As it passes through position **a1** the counter updates and the **a1** memory is activated

A-

Once the piston is fully extended the retraction phase starts (***A-***)

If the **start memory** stays **on** the cycles restarts from the extension phase

If the **memory** turns **off** or the emergency is pressed the cycle is brought back to the initial step



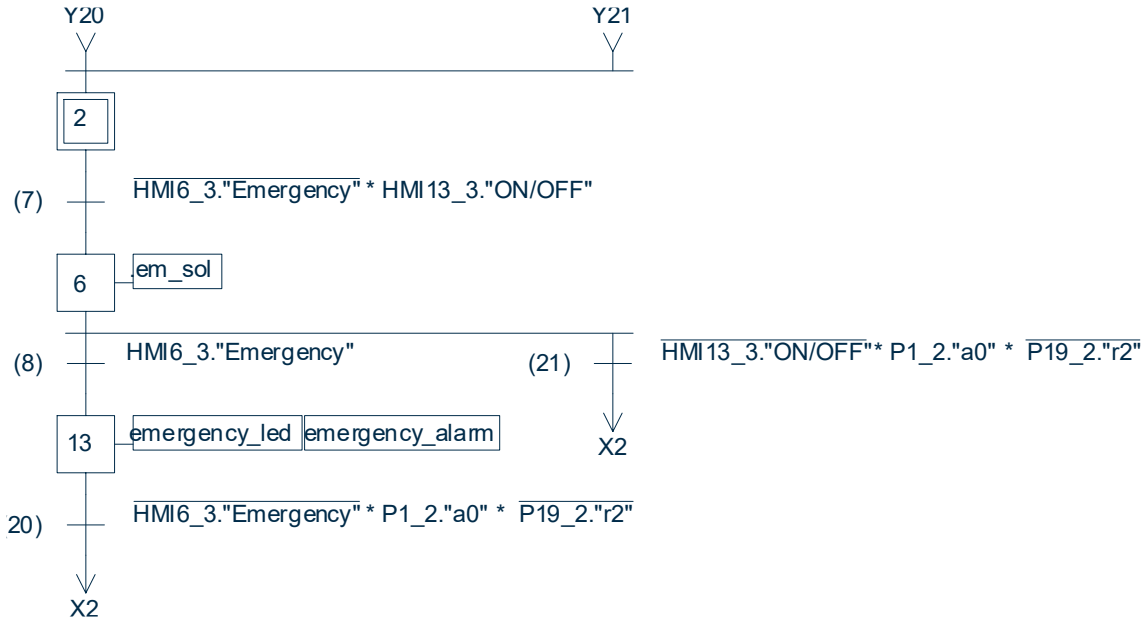
Grafcet

Emergency

If the **emergency twist** and **release button** is not pressed the emergency solenoids are activated

LED and the **alarm** turn **on**

Devices stay **on** until the push button is **released** and the **homing** is performed

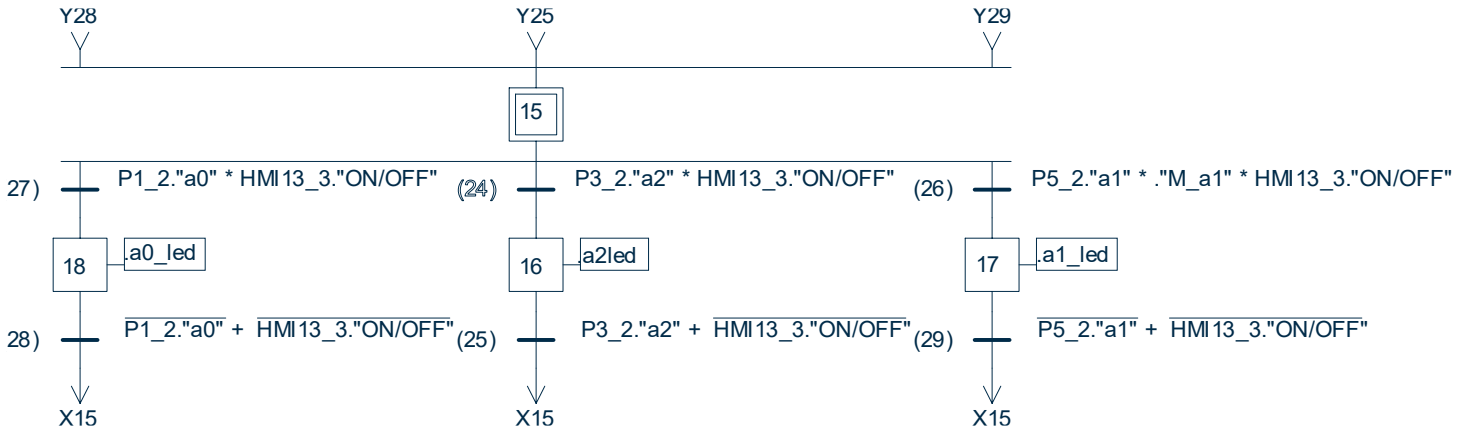


Grafcet

LED

When the **ON/OFF switch** is **on**, the **a0**, **a1** and **a2 LEDs** are switched **on**

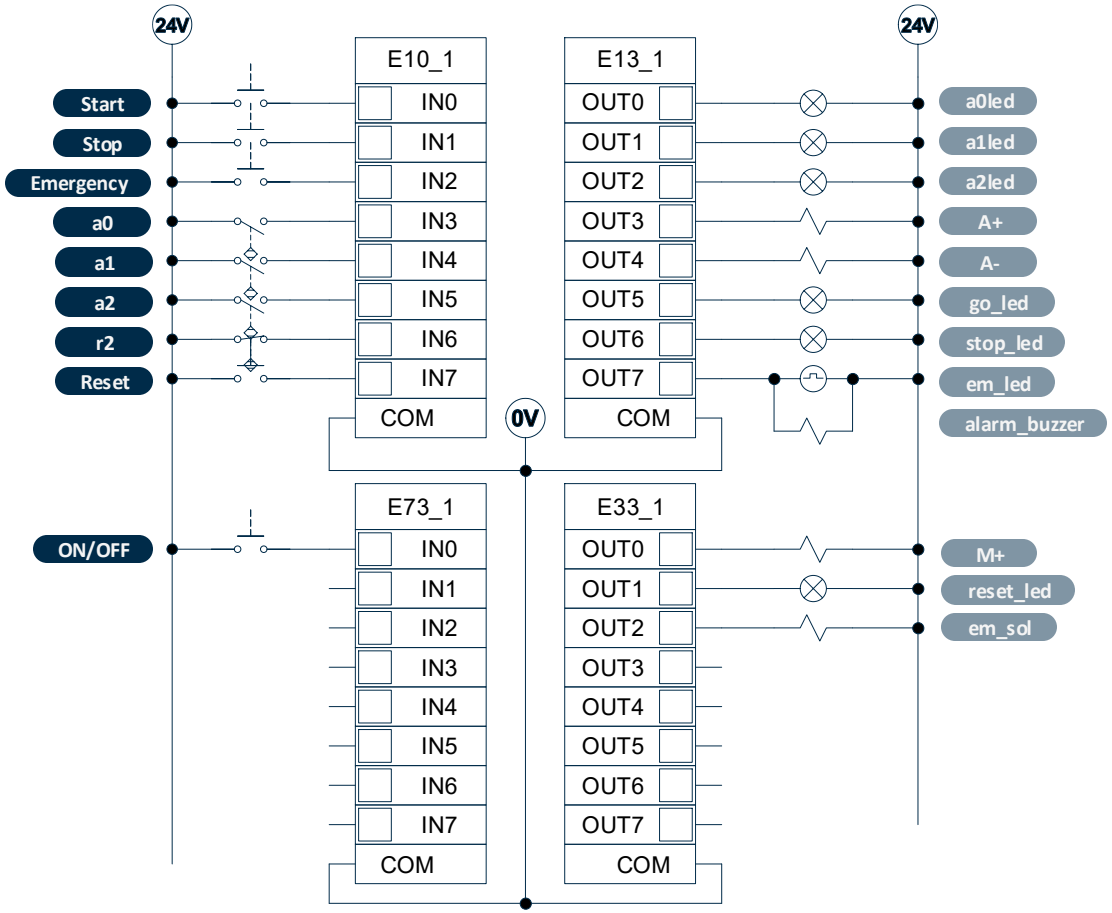
a1 LED only toggles when the **a1** memory is **active**



PLC

input connections

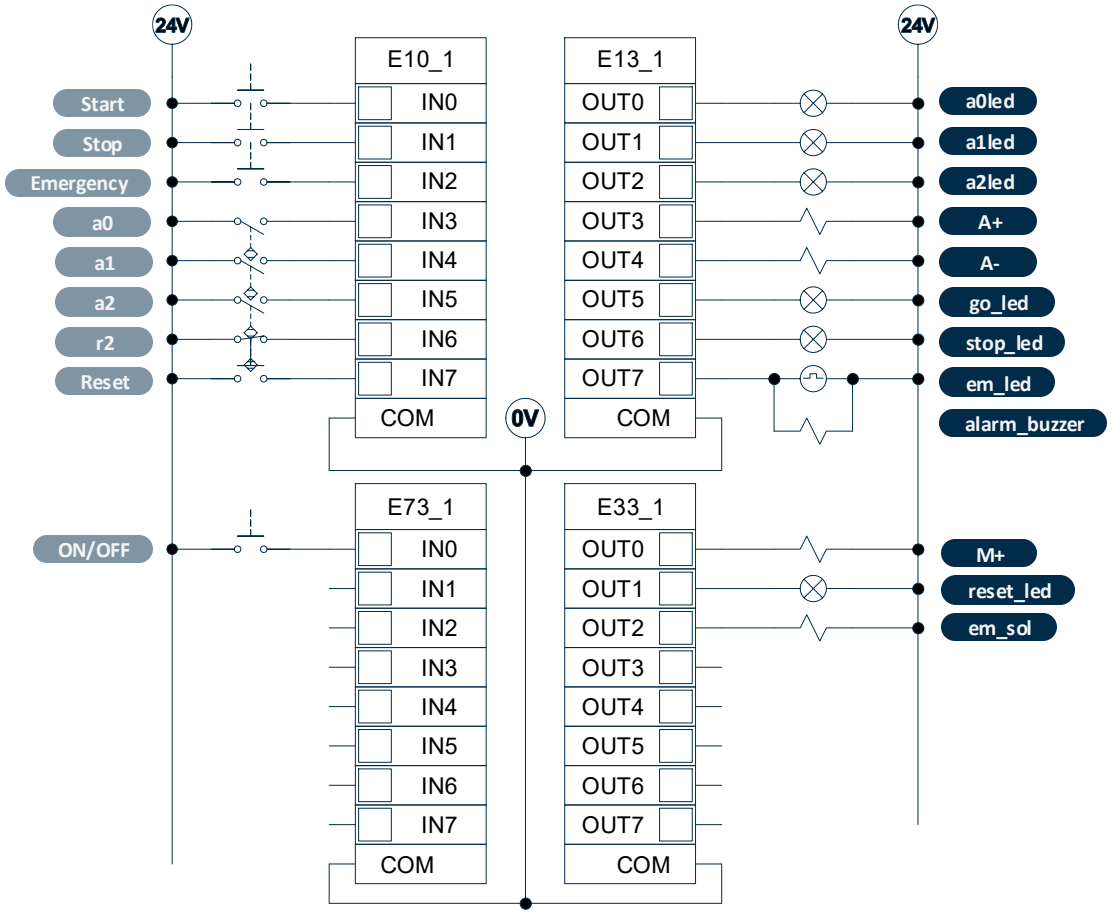
	component	functionality
E10_1.IN0	pushbutton	Start
E10_1.IN1	pushbutton	Stop
E10_1.IN2	pushbutton	Emergency
E10_1.IN3	switch	a0
E10_1.IN4	switch	a1
E10_1.IN5	switch	a2
E10_1.IN6	switch	r2
E10_1.IN7	pushbutton	Reset
E73_1.IN0	pushbutton	ON/OFF



PLC

output connections

	component	functionality
E13_1.OUT0	LED	a0led
E13_1.OUT1	LED	a1led
E13_1.OUT2	LED	a2led
E13_1.OUT3	solenoid	A+
E13_1.OUT4	solenoid	A-
E13_1.OUT5	LED	go_led
E13_1.OUT6	LED	stop_led
E13_1.OUT7	coil	em_led
E13_1.OUT7	solenoid	alarm_buzzer
E33_1.OUT0	solenoid	M+
E33_1.OUT1	LED	reset_led
E33_1.OUT2	solenoid	em_sol



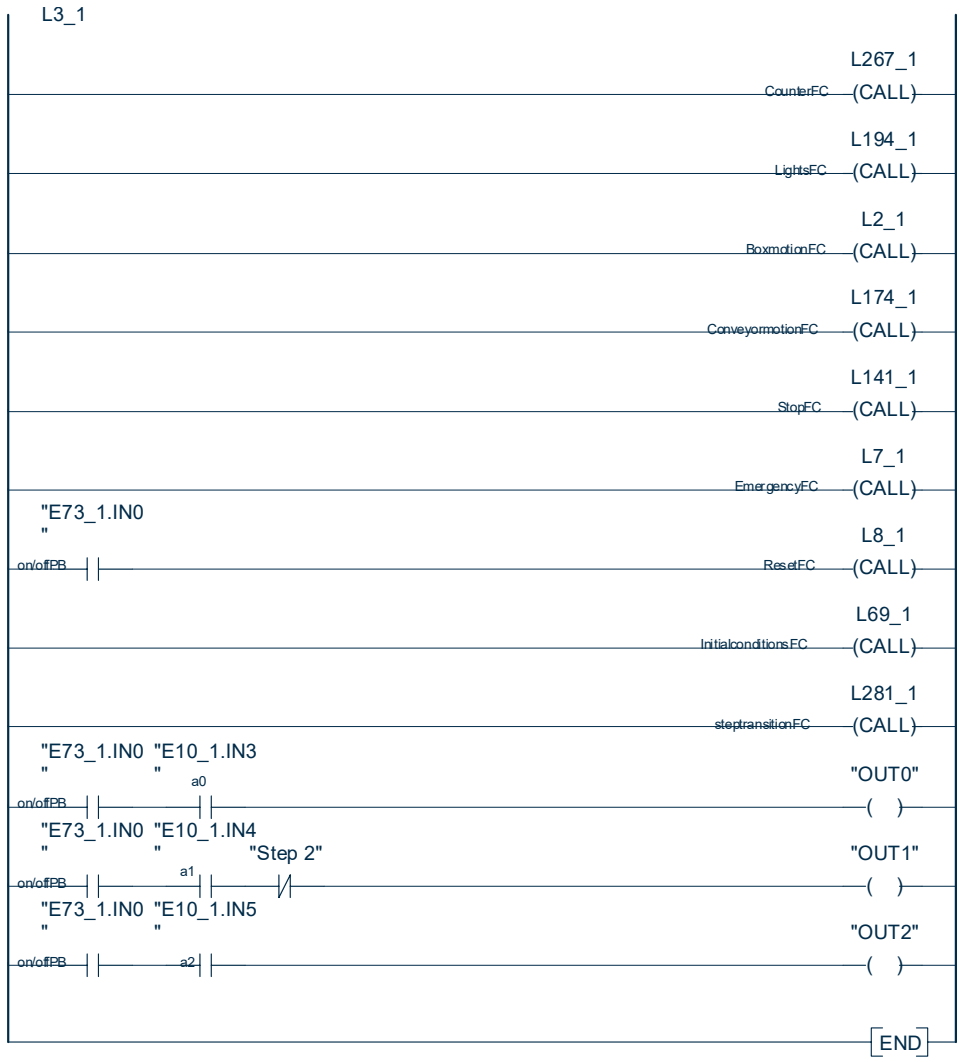
PLC

main

The main rung contains the **calls** to all the following function

The main rung contains the **logic** for the switching of the position LEDs for the piston used to simulate the motion of the box

The latter **not** being displayed in the HMI were **not** included in the lights function



PLC

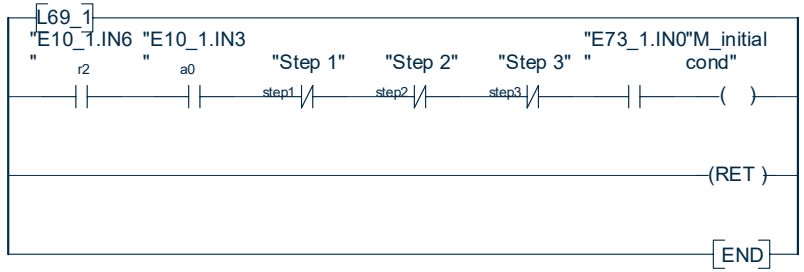
Initial condition FC

a0

The initial condition function controls the **position** of the piston rod (**a0**)

M

The presence of an air flow to the pneumatic **motor** (**r2**)



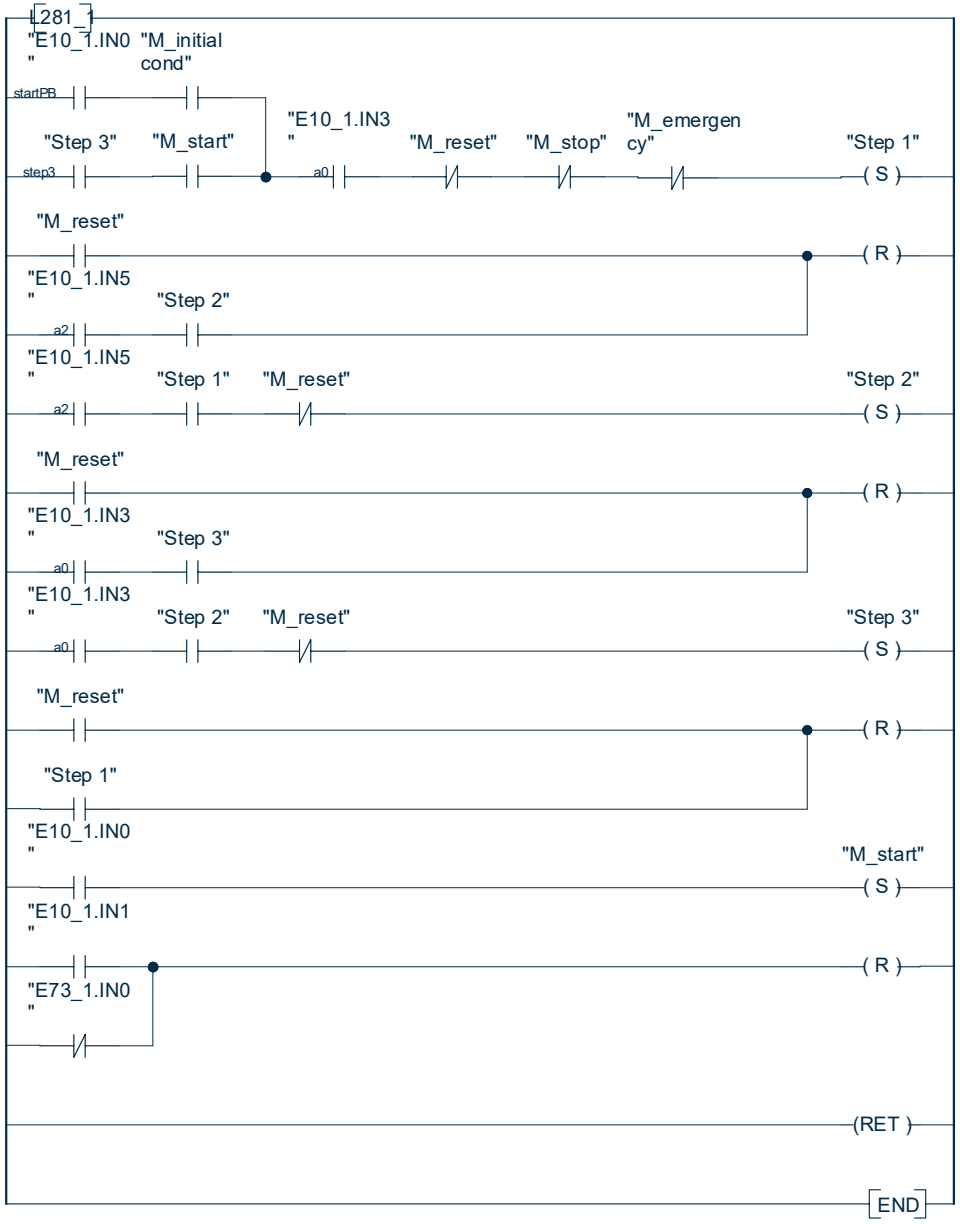
The **resetting** of all the step 1 to 3 used inside the *Step transition FC* and that the machine has been turned ON (ON/OFF*)*

PLC

Step transition FC



The step transition function controls the logic of the motion of the **piston rod**



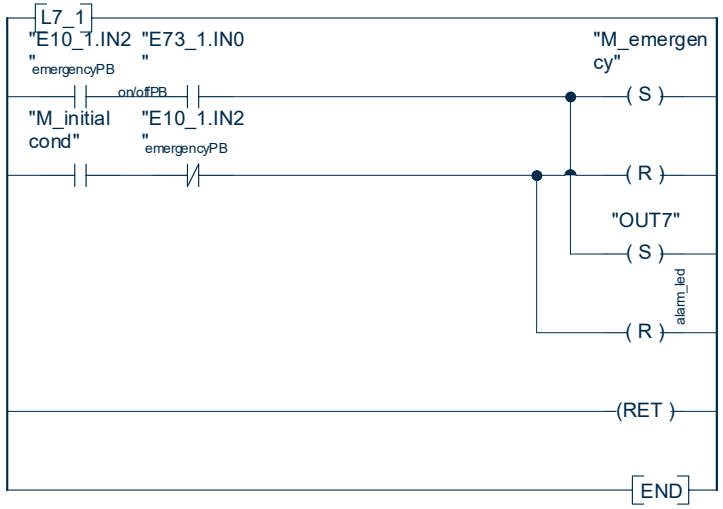
PLC

Emergency FC

The **emergency** function activates when the **twist release button** is pressed

M

It sets the emergency memory (**M_emergency**) and power the solenoid that controls the alarm and the emergency LEDs (**OUT_7**)



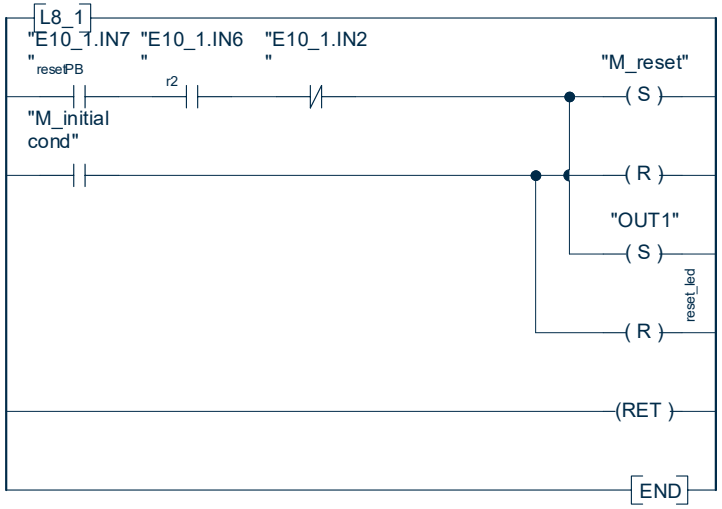
PLC

Reset FC

The **reset** function activates only when the **reset push button** is pressed, the **emergency button** is not pressed, and the **pneumatic motor** is not running (system **stopped**)

Upon **activation** the **reset** memory is set as well as the reset **LEDs**

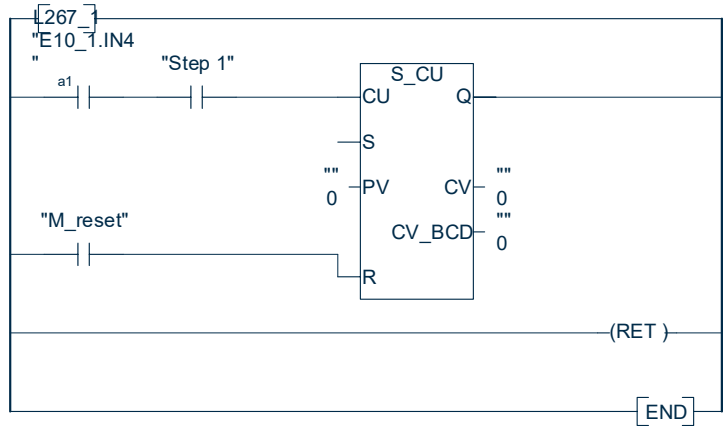
As soon as the **initial conditions** are reached all the previous memories are reset



PLC

Counter FC

The **counter** function activates only when the **a1** contact closes and the **step 1** is set, to prevent counting in both outward and inward stroke of the piston



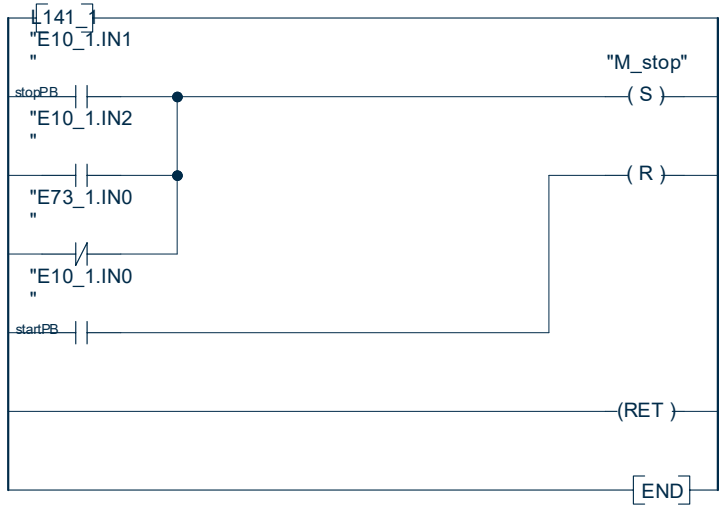
It is **reset** when the reset memory (**M_reset**) is **set**

PLC

Stop FC

The **stop function** sets the stop memory (**M_stop**) when the **stop push button** is pressed, when the **emergency** button is pressed or when the **ON/OFF switch** is OFF

As soon as the **start push button** is pressed the stop memory is **reset**



PLC

Box motion FC

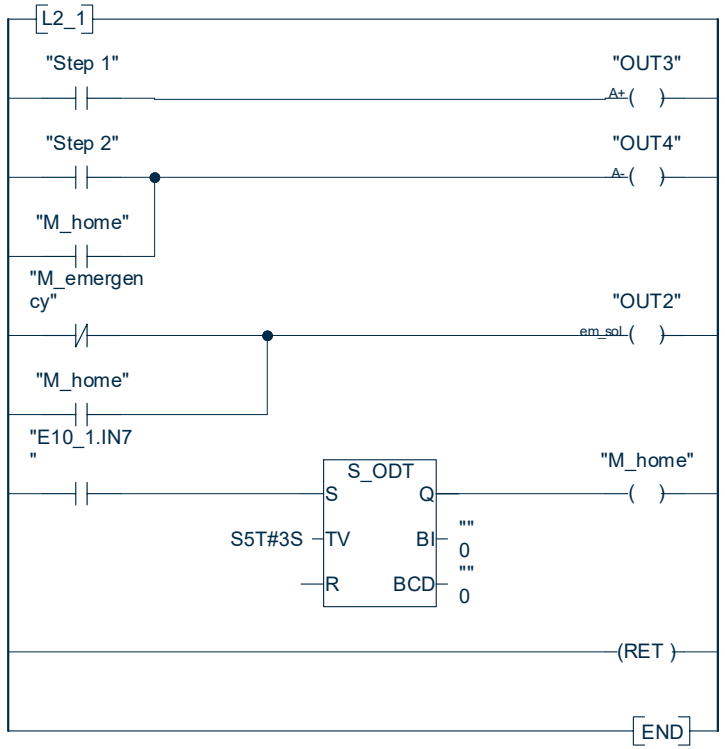
The **box motion** function activates the outward and inward strokes when the **step1** and **step 2** are set, respectively

It sets the **emergency** solenoids unless the emergency memory is set

When the **reset** push button is pressed for more than **3s** the **home memory** is set (*M_home*)

HOLD 3s

In this situation the **emergency solenoids** are activated as well as the **A- solenoid** to allow the **homing** of the piston



PLC

Conveyor motion FC

M+

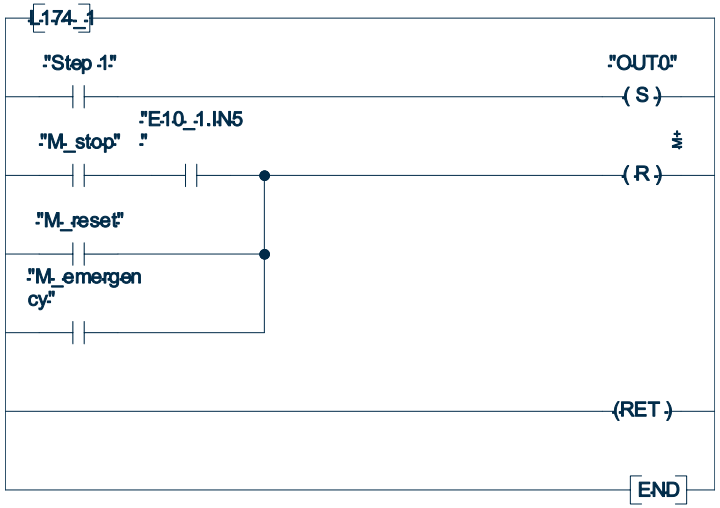
The **conveyor motion** function activates the **M+ solenoid** as soon as the cycle enters the **step 1**

M+

The solenoid is **deactivated** when the **emergency, reset or stop memory** are set

a2

The latter also requires the activation of **a2** to ensure the box reaches the end of the conveyor before stopping the pneumatic motor



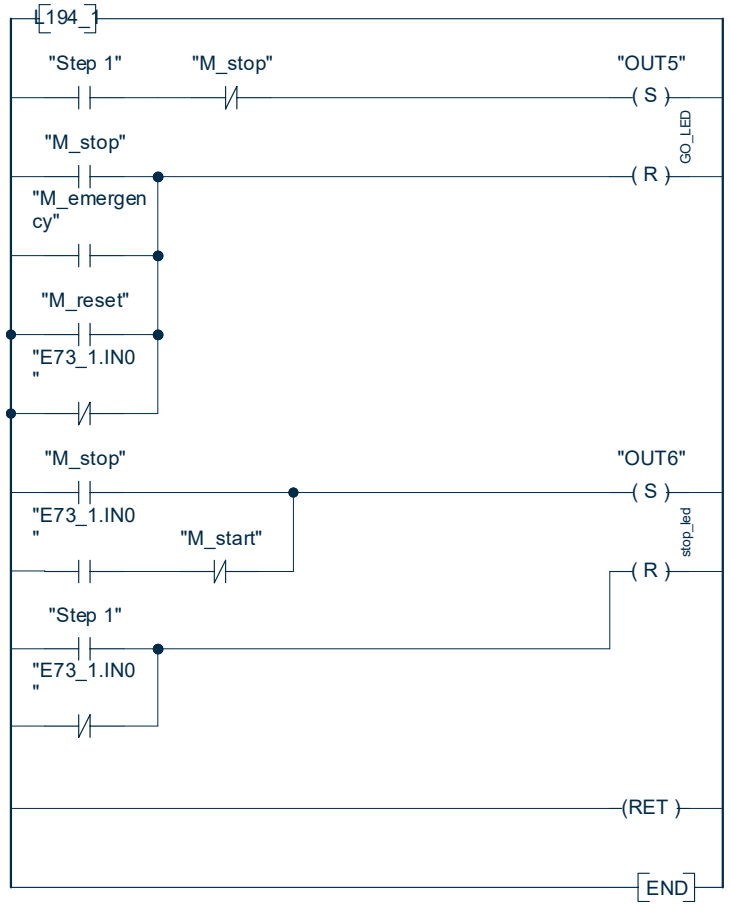
PLC

Lights FC

The **lights** function activates the **running** and the **stop** LEDs

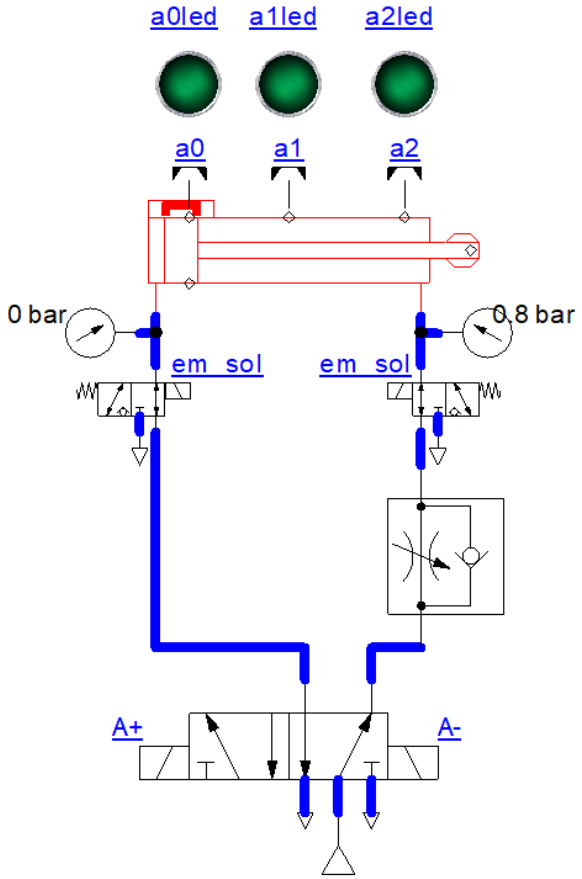
When the machine **starts** the running LEDs (*GO_LED*) are **set** until the **emergency, stop or reset memories** are **activated**, or the machine is turned **off**

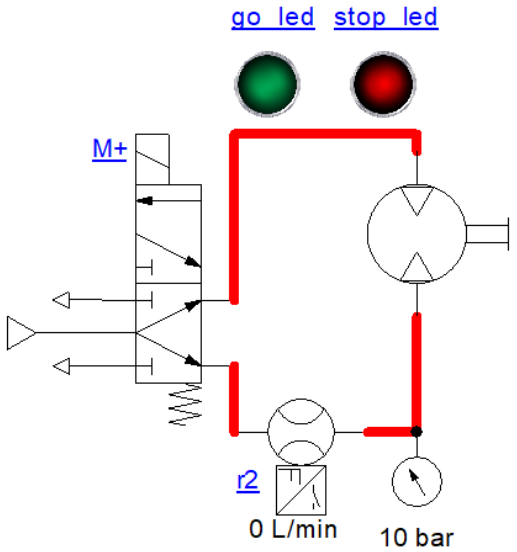
On the other side when the machine **stops** the stop LEDs (*stop_led*) are set until the machine is **restarted** or **switched off**



Simulation

overview



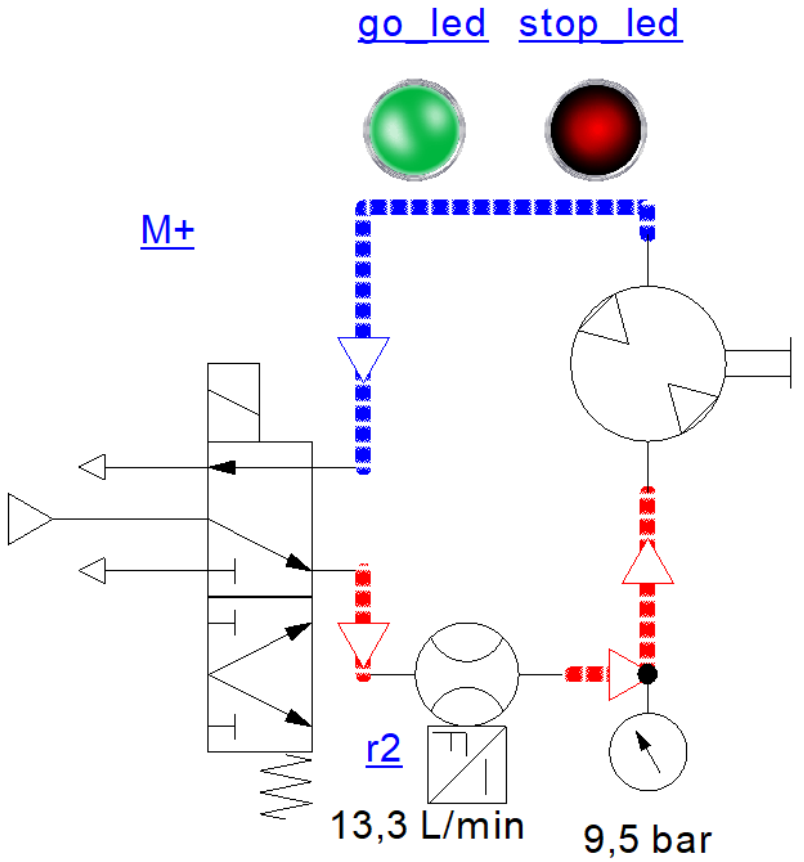


Simulation

Start

When the **start** pushbutton is pressed the **motor** starts and the **box** is moving along the belt

The **go_led** is on and the **counter** is ready for counting when the **box** passes in front of the photosensor

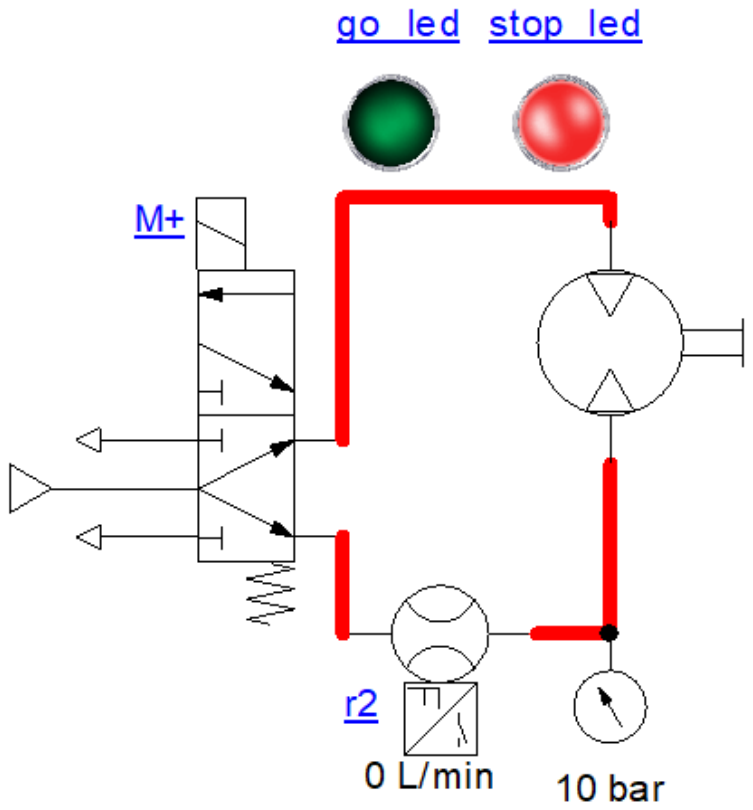


Simulation

Stop

When **stop** button is pressed or the machine is turned **ON** the motor is **stopped** and the box is at the start of the belt

When the belt is stopped a **stop_led** is on

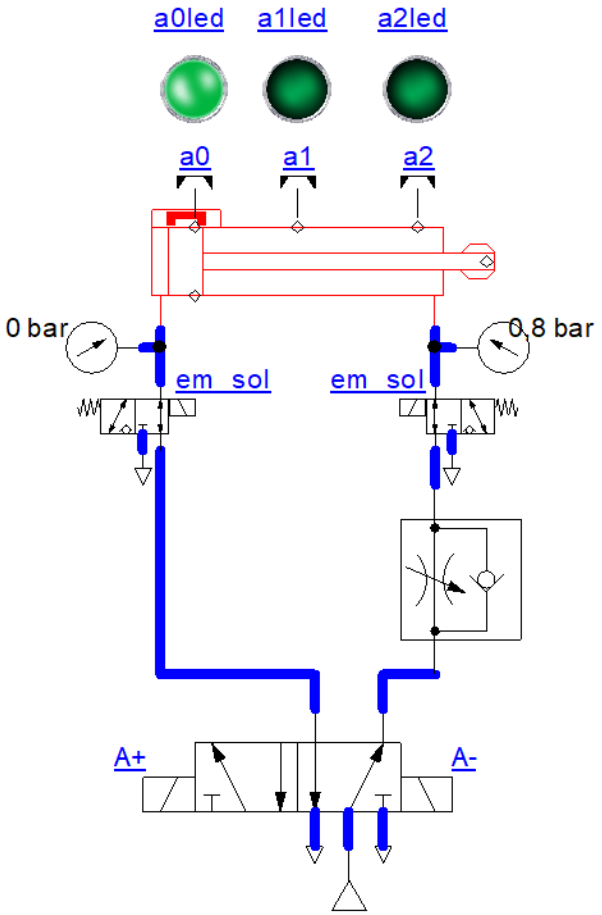


Simulation

a0

When the **stop** pushbutton is pressed the cycle is completed and when the piston arrives in position **a0** it stops

The **a0 led** stay on till the machine is turned **OFF**

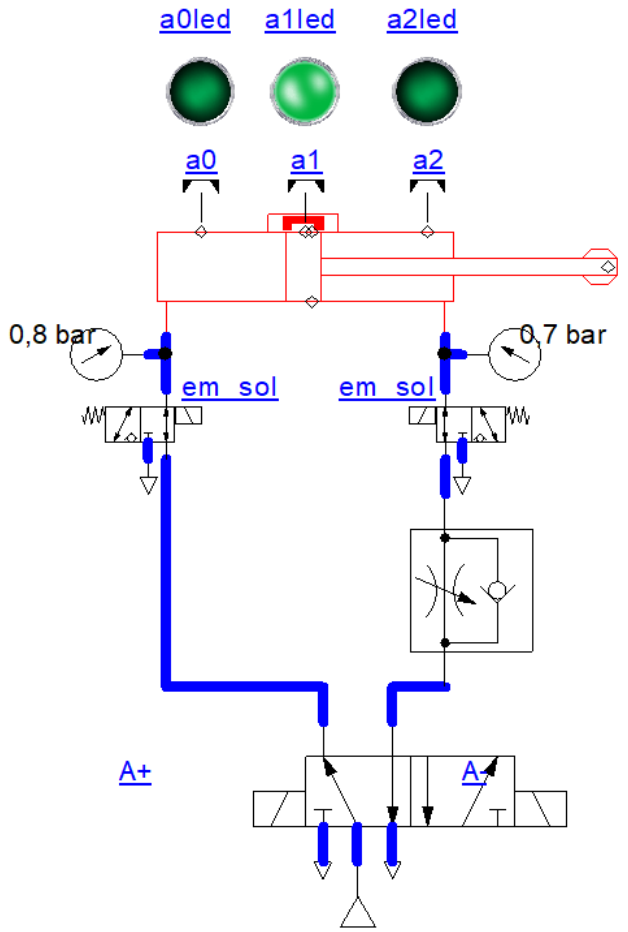


Simulation

a1



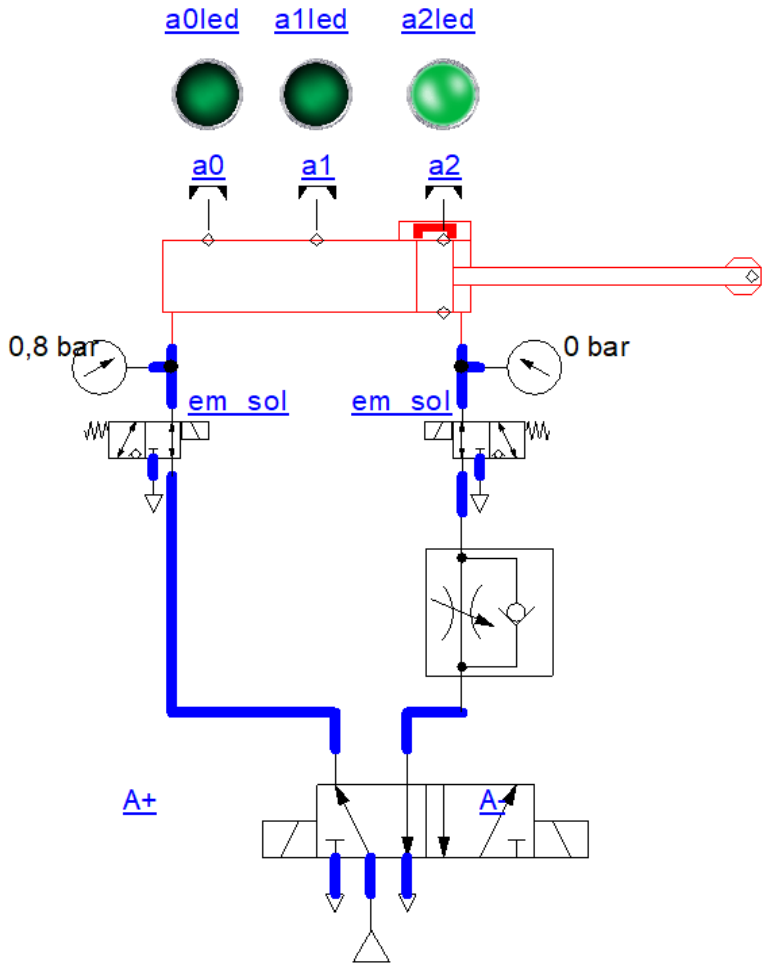
When the **start** pushbutton is pressed the piston starts its **outstroke** movement and a **led** is turned on every time the piston passes in position **a0**, **a1** or **a2**



Simulation

a2

When the **start** pushbutton is pressed the piston starts its **outstroke** movement and a **led** is turned on every time the piston passes in position **a0**, **a1** or **a2**

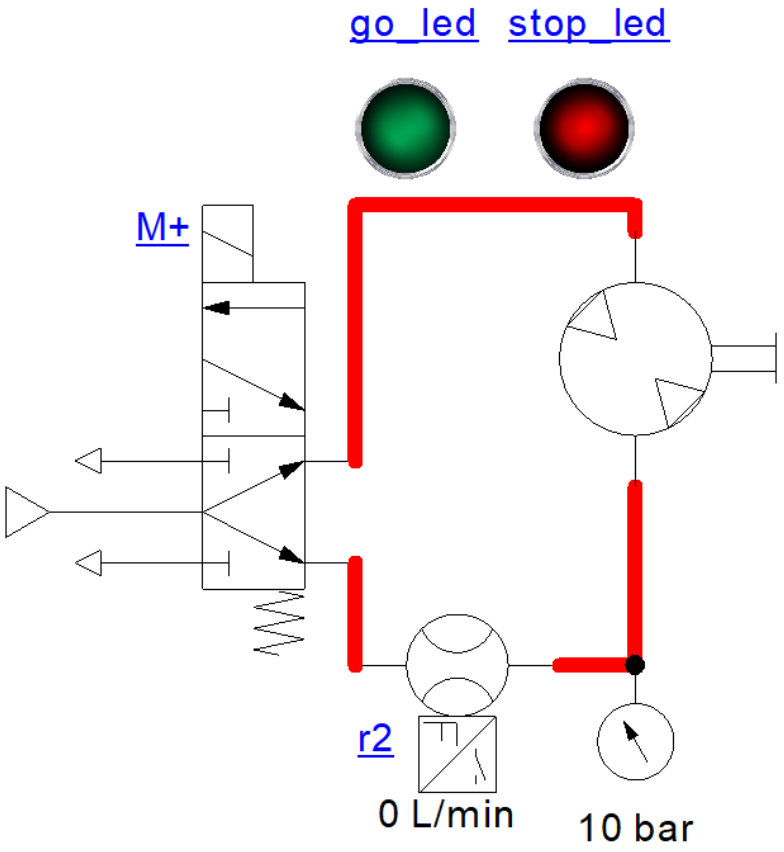


Simulation

emergency

When the **emergency** button is pressed the belt is **stopped** immediately

During this time an **emergency_led** is **on** and an **alarm buzzer** is ringing

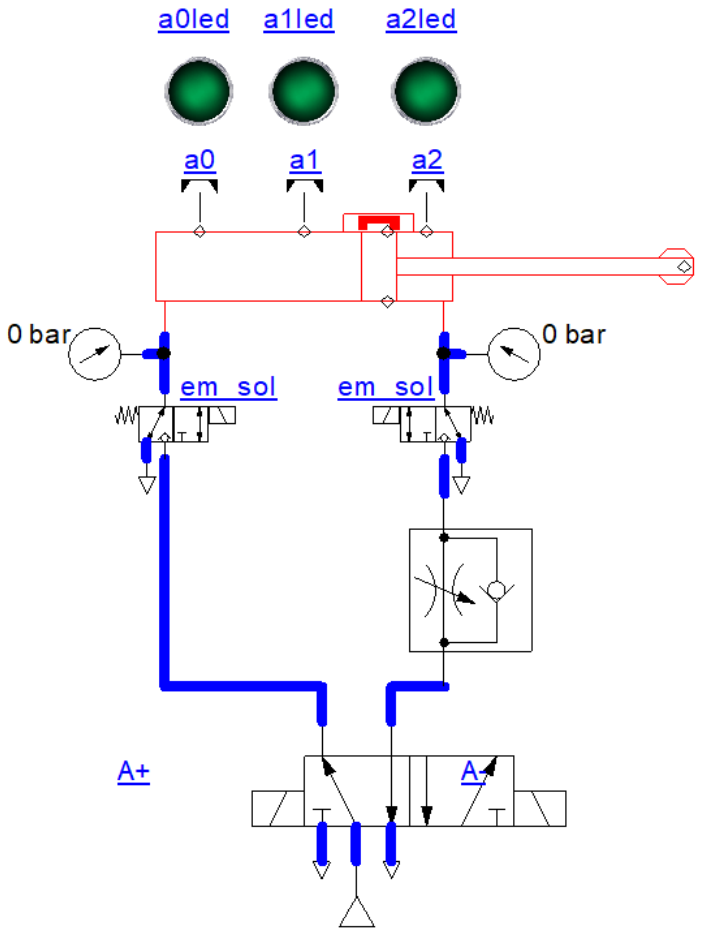


Simulation

cylinder emergency

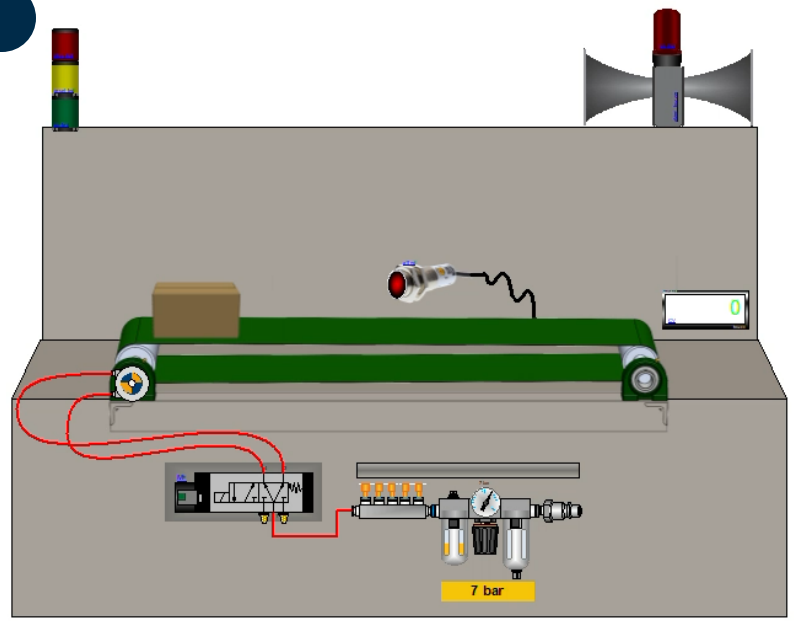
When the **emergency** button is pressed the belt is **stopped** immediately

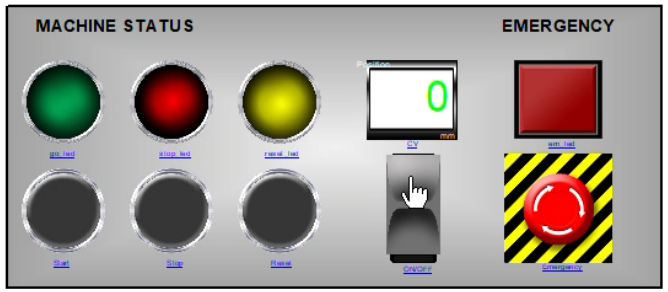
During this time an **emergency_led** is **on** and an **alarm buzzer** is ringing



Simulation

video





Results

state diagrams

ON/OFF

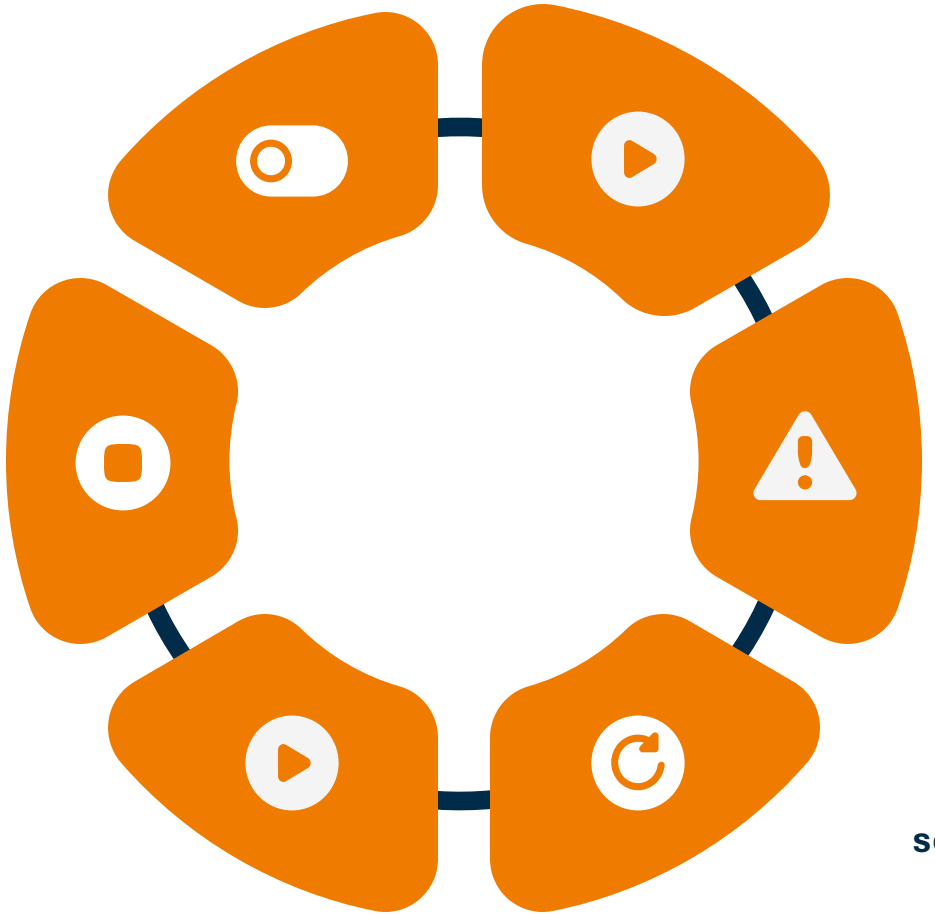
the machine is turned **ON/OFF**

Stop

the machine **stops** after the current cycle has ended

Start

the cycle **starts** and the counter counts the boxes



Start

the cycle **starts** and the counter counts the boxes

Emergency

an **emergency** stop is performed, and the cycle is interrupted

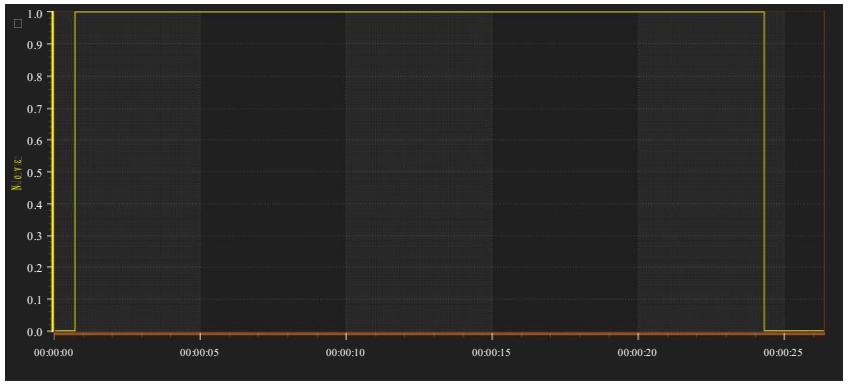
Reset

the **reset button** has been pressed for **3 seconds** to perform the homing of the piston and reset the value of the counter

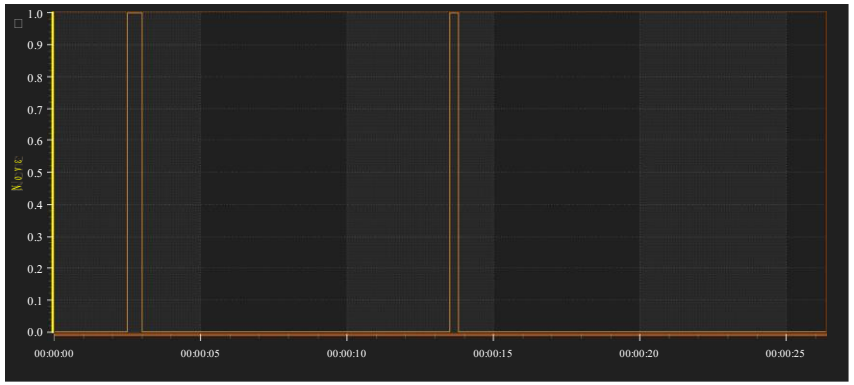
Results

system commands

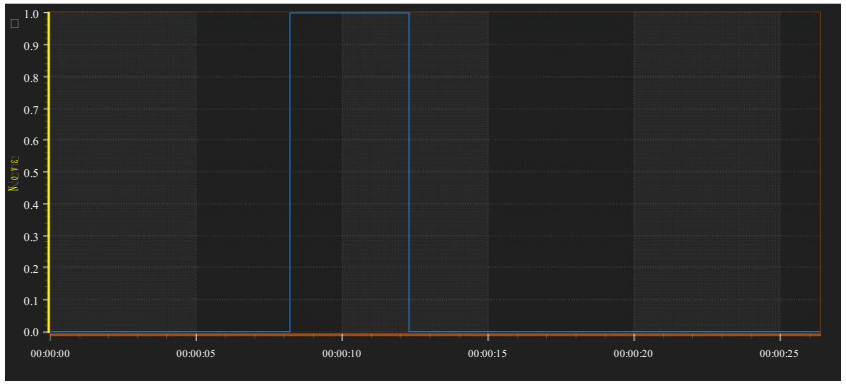
ON/OFF



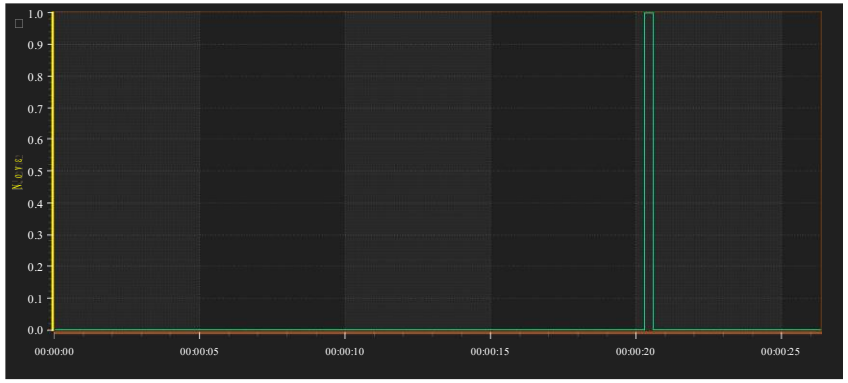
Start



Reset



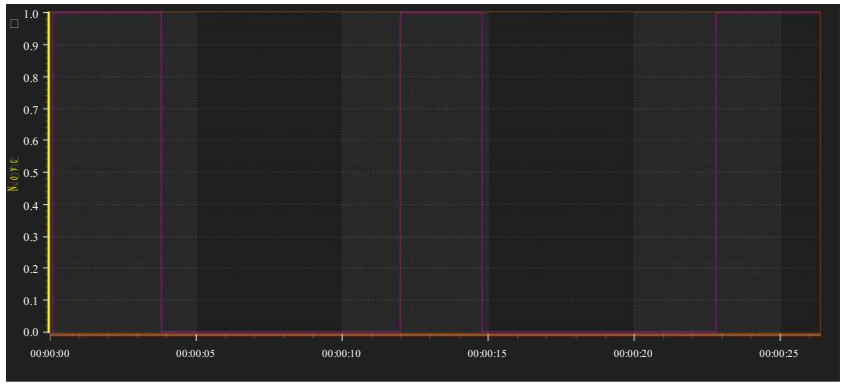
Stop



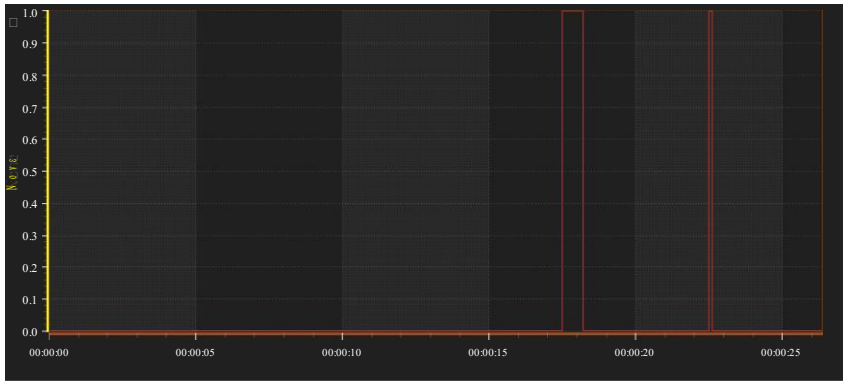
Results

system sensors

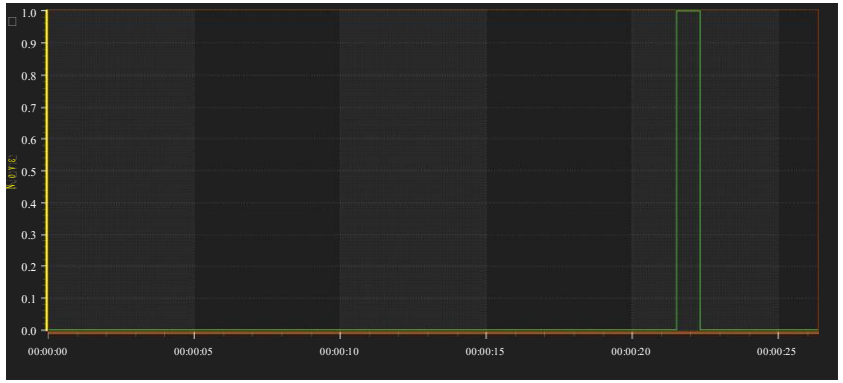
a0



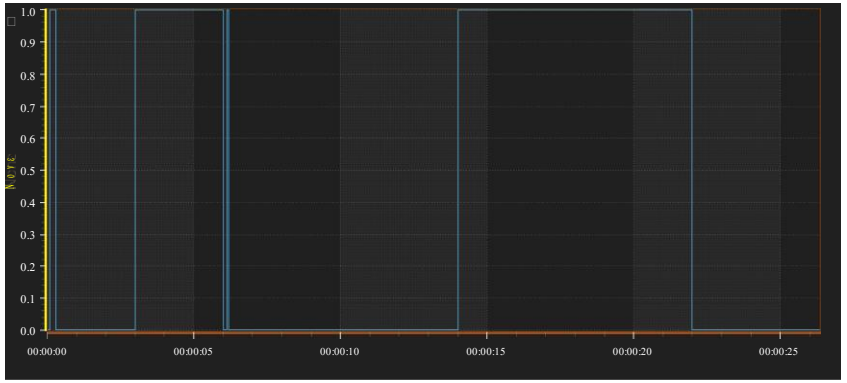
a1



a2



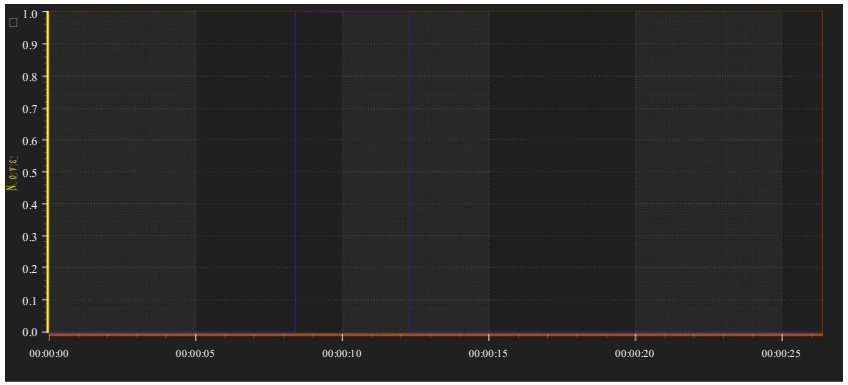
Flow sensor



Results commands LEDs

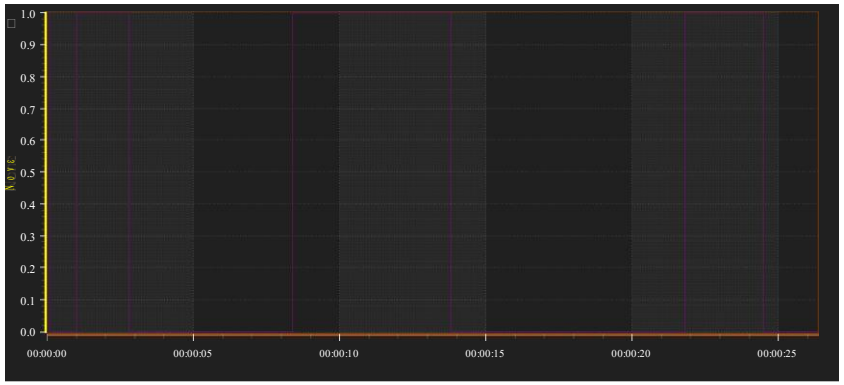
Go LED





Reset LED

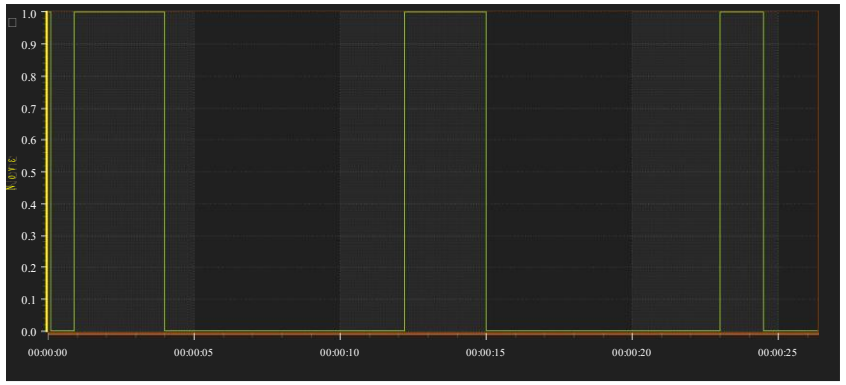
Stop LED

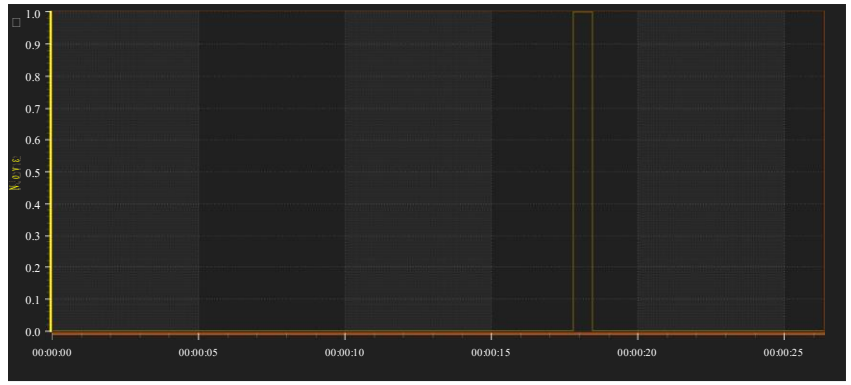


Results

sensors LEDs

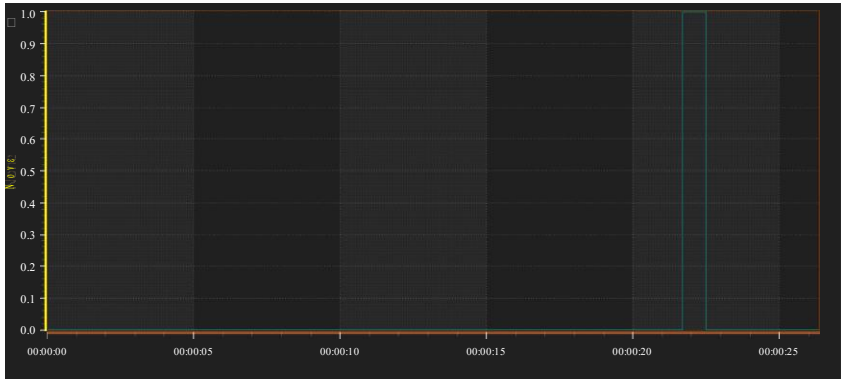
a0 LED





a1 LED

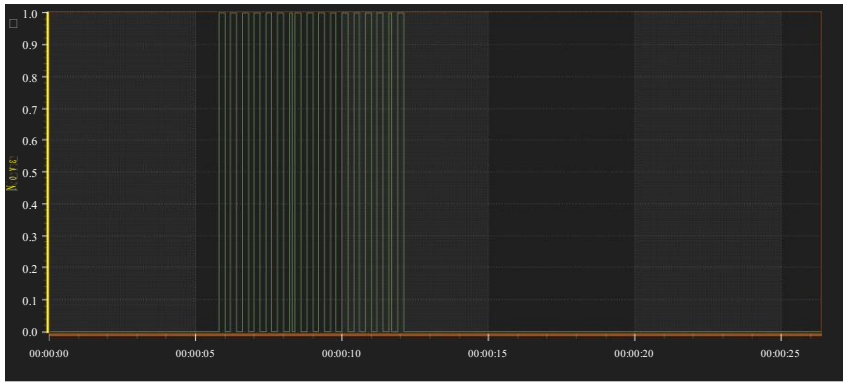
a2 LED

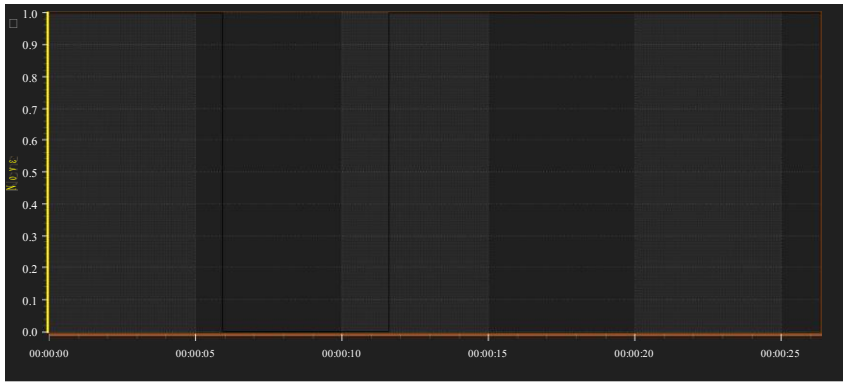


Results

emergency system

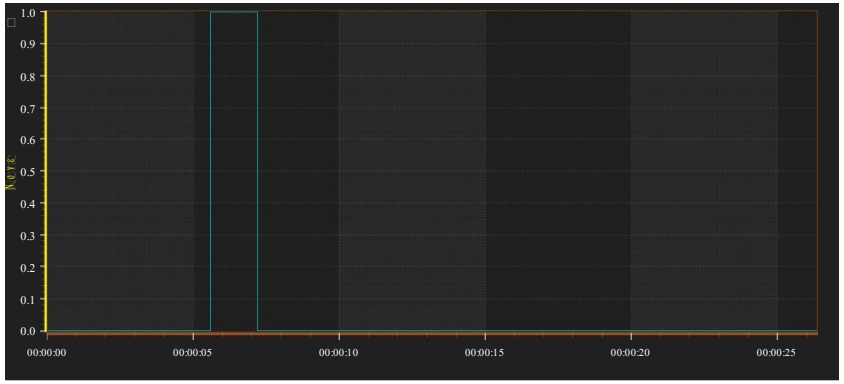
LED





Solenoid

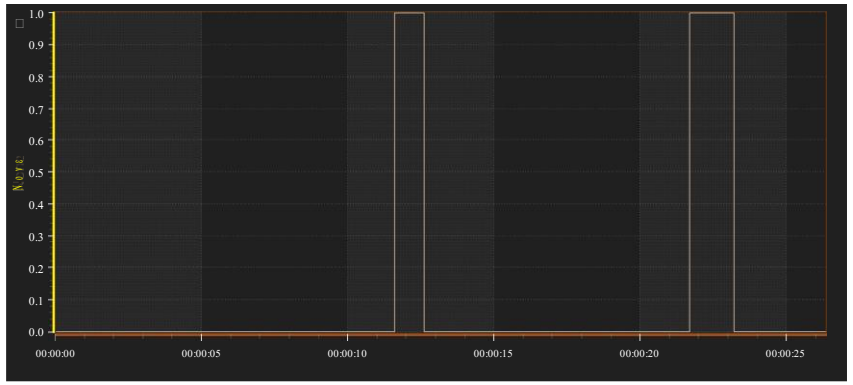
Twist-release



Results solenoids

Cylind. ext.





Cylind. retr.

Mot. activ.

